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0178/01

October/November 2015

1 hour

Additional Materials: Geometrical Instruments
Tracing Paper (optional)

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

If work is needed for any question it must be shown below that question.

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER.

The number of marks is given in brackets [] at the end of each question or part question.
The total of the marks for this paper is 60.

1 Work out the following.

(a) $3 + 2 \times 7 - 4$

Answer (a) [1]

(b) $2\frac{2}{3} - 1\frac{1}{4}$

Answer (b) [2]

(c) 3^{-2}

Answer (c) [1]

2 From the following list of numbers

$\sqrt{3}$ π 2 32 35 121

write down

(a) a multiple of 7,

Answer (a) [1]

(b) a factor of 8,

Answer (b) [1]

(c) a prime number,

Answer (c) [1]

(d) a square number,

Answer (d) [1]

(e) an irrational number.

Answer (e) [1]

3 Write the order of rotational symmetry of the shape drawn.



Answer [1]

- 4 The population of Lesotho is approximately 1 870 000.

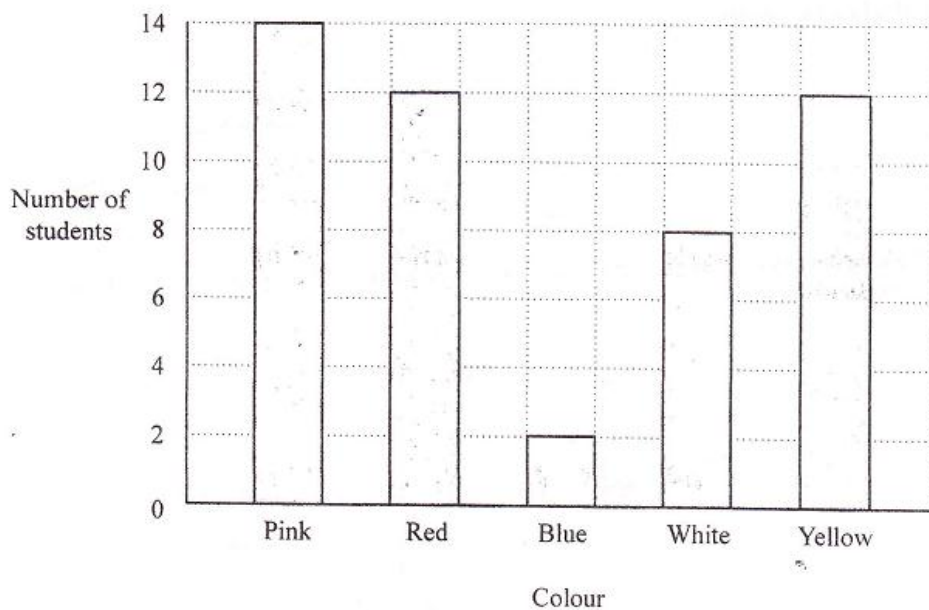
(a) Write 1 870 000 to the nearest million.

Answer (a) [1]

(b) Write 1 870 000 in standard form.

Answer (b) [1]

- 5 In a Form A class, students were asked to indicate their favourite colours.
The results are shown in the bar chart.



(a) How many students are there in the Form A class?

Answer (a) [2]

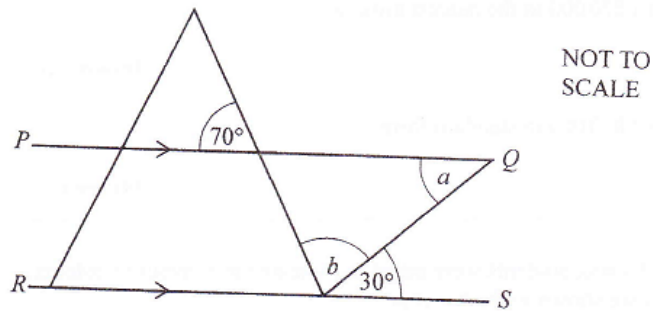
(b) Which colour is liked best by the students?

Answer (b) [1]

(c) Find the fraction of students whose favourite colour is white.

Answer (c) [1]

- 6 In the diagram, PQ and RS are parallel lines.

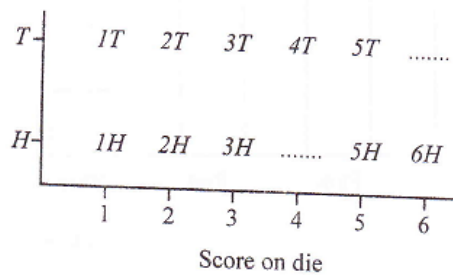


Calculate the angles a and b .

Answer $a =$

$b =$ [3]

- 7 The diagram shows the possible outcomes when a die is thrown and a coin is tossed.
(T - tails, H - heads)



- (a) Complete the diagram.

[1]

- (b) Find the probability of obtaining

- (i) a tail and an odd number less than 4,

Answer (b)(i) [1]

- (ii) a number less than three with a head or tail.

Answer (b)(ii) [1]

- 8 When 20 kg of carrots are peeled, 1 kg is lost as peel.

Find this percentage loss.

Answer% [2]

- 9 The exchange rate from Maluti (M) to Pula (P) is P 1.00 = M 1.07.

Molise travels from Botswana to Lesotho.

He has P 2000 to exchange into Maluti.

How much Maluti does he receive?

Answer M [2]

- 10 Factorise completely.

$$2xy + 6xz$$

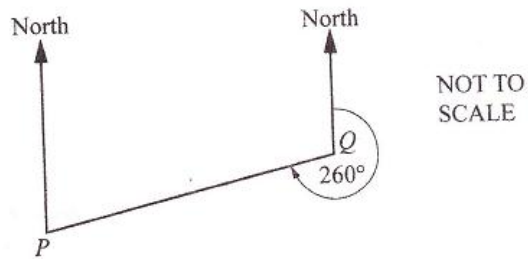
Answer [2]

- 11 Solve the equation.

$$\frac{y+1}{5} = 2$$

Answer $y =$ [2]

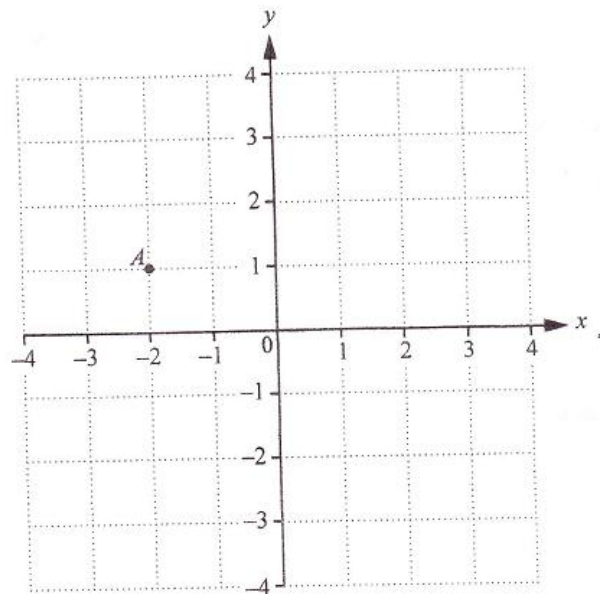
- 12 In the diagram, the bearing of P from Q is 260° .



Find the bearing of Q from P .

Answer [2]

13



- (a) Plot the point $B(2, -2)$. [1]

- (b) Find the gradient of the line AB .

Answer (b) [2]

- (c) Write down the column vector $\overrightarrow{AB}$.

Answer (c) $\begin{pmatrix} \\ \end{pmatrix}$ [1]

- 14 The table shows the shoe sizes of a group of students.

Shoe size	Number of students
3	7
4	14
5	12
6	3
7	3
8	1

- (a) How many students are in the group?

Answer (a) [1]

- (b) Find the median shoe size.

Answer (b) [2]

- 15 $f(x) = 2x + 3$

Find $f(2)$.

Answer [1]

- 16 Arrange the following in order of size, starting with the smallest.

$$\frac{5}{12} \quad 0.25 \quad \frac{1}{6} \quad \frac{2}{3}$$

Answer < < < [2]

- 17 The picture is drawn to the scale of 1 : 250.

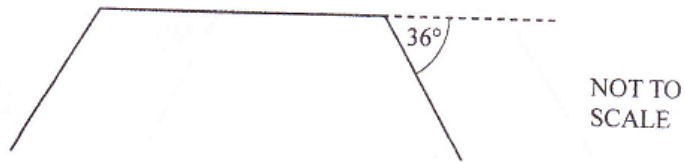


The height of the tree in the picture is 3 cm.

Find the actual height of the tree in metres.

Answer m [2]

- 18 The diagram shows part of a regular polygon.



Calculate the number of sides of the polygon.

Answer [2]

- 19 Solve the simultaneous equations.

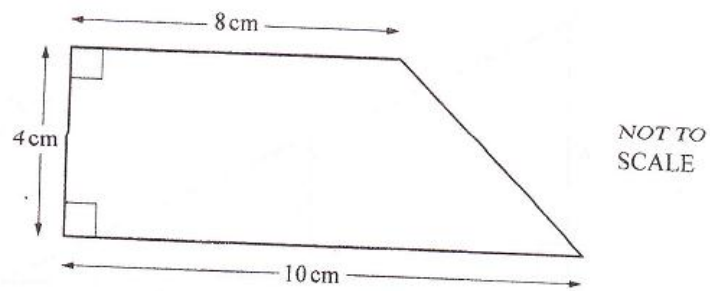
$$x + y = 7$$

$$5x + 2y = 20$$

Answer $x =$

$y =$ [3]

- 20 The diagram shows a trapezium.



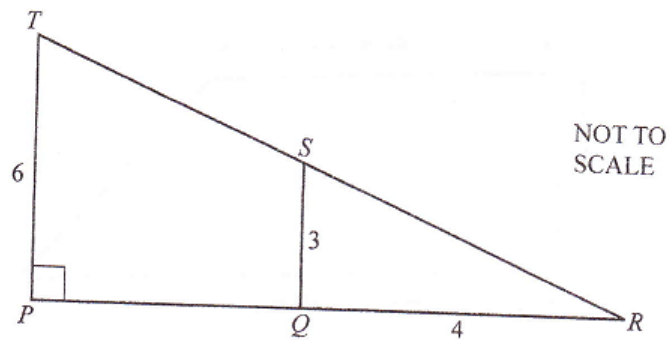
Calculate the area of the trapezium.

Answer cm^2 [3]

- 21 The distance, d , from home to school is 850 metres correct to the nearest 10 m.
Find the upper and the lower bound of the distance.

Answer $\text{m} \leq d < \dots \text{m}$ [2]

- 22 In the diagram, triangle PRT is similar to triangle QRS .

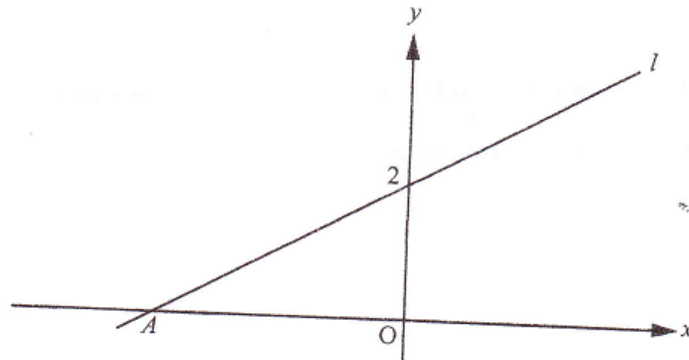


$QS = 3$ cm, $QR = 4$ cm and $PT = 6$ cm.

Find the length PR .

Answer cm [2]

23



The line l has a gradient of $\frac{1}{2}$.

- (a) Write down the equation of l .

Answer (a) $y = \dots\dots\dots$ [2]

- (b) The line l meets the x -axis at A .

Find the x -coordinate of A .

Answer (b) $(\dots\dots\dots, 0)$ [2]